

Grade 6  
Unit 8  
Lesson 8 Practice

Name \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

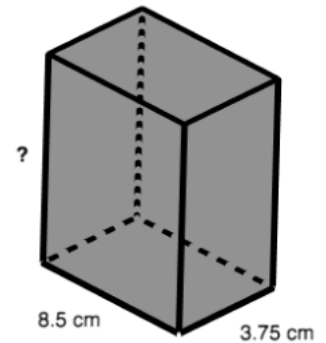
A rectangular prism is shown with measurements in centimeters (cm).

1. What is the area of the bottom face of this prism? Include units.

$$8.5 \times 3.75 = 31.875 \text{ cm}^2$$

2. If the volume of the box is 382.5 cubic centimeters ( $\text{cm}^3$ ), what is the value of the missing dimension of the box?

$$382.5 = 31.875 \cdot h \Rightarrow h = 12 \text{ cm}.$$



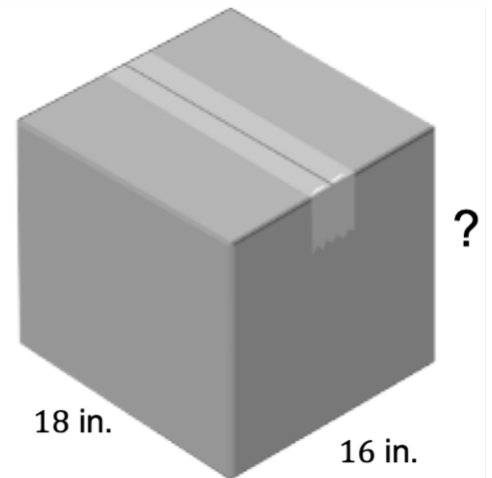
Sarai is using moving boxes from the post office. The boxes have a length of 18 inches (in.) and a width of 16 in., but she cannot recall the height of the box

3. Determine the area of the bottom face of the box. Include units.

$$18 \times 16 = 288 \text{ in}^2$$

4. If the volume of the box is 5,184 cubic inches ( $\text{in}^3$ ), what is the height of the box? Include units.

$$5184 = 288h \Rightarrow h = 18 \text{ in}$$



Jenni was trying to find the missing dimension of the lunchbox shown. She made a mistake when solving for the width of the box, but she cannot find it. Her work is shown.

Volume = 234 cubic in.



| Jenni's work                                                       |                      |
|--------------------------------------------------------------------|----------------------|
| $\begin{array}{r} 6.5 \\ \times \quad 9 \\ \hline 585 \end{array}$ | $234 \div 5.85 = 40$ |
| $585 \rightarrow 5.85 \text{ in}^2$                                | $w = 40 \text{ in.}$ |

5. Circle the mistake.
6. What should the width of the box be? Include units.

$$234 \div 58.5 = 4 \Rightarrow w = 4 \text{ in.}$$

7. Write a note to Jenni on how to correct her error(s).  
 The decimal point should be moved one decimal place to the left.  
 $585 \Rightarrow 58.5$

8. A pizza box has a length of 14 inches (in.) and a height of 2.25 in. The box has a volume of 441 cubic inches ( $\text{in}^3$ ). What is the width of the box? Include units.

$$441 = (14 \cdot 2 \cdot 25) \cdot w$$

$$441 = 31.5 \cdot w \Rightarrow w = 14 \text{ in.}$$

9. Mike bought an aquarium. He knows that the length is 36 inches (in) and the width is 18.25 in., but he can't remember the appropriate height. The volume needs to be 11,169 cubic inches ( $\text{in}^3$ ). What is the height of the aquarium? Include units.

$$11,169 = (36 \cdot 18.25) \cdot h$$

$$11,169 = 657h \Rightarrow h = 17 \text{ in.}$$

10. Sharee is measuring her pencil case. The case has a width of 4.25 inches (in.) and a height of 2 in. The volume of the case is 68 cubic inches ( $\text{in}^3$ ). Determine the length of the suitcase. Include units.

$$68 = (4.25 \cdot 2) \cdot \ell$$

$$68 = 8.5 \cdot \ell$$

$$8 = \ell \Rightarrow 8 \text{ in.}$$

